ANTENATAL CARE (A.N.C.)

Definition:

It's a program of preventive obstetrics defined to detect the abnormalities or impending abnormalities of pregnancy at early stage & to correct them, and to educate the mother the sound habits during pregnancy.

So we do screening of the entire pregnant population to:

- Detect women at high risk. High risk preg. (HRP) referring them for specialized obstetric care
- Follow-up the low risk pregnancy. (LRP) through out pregnancy till labour.

Components (Technique) of A.N.C:

(A)<u>Health education</u>: Usually instruction for the patient in 1st visit (Booking visit):

I- Life style during pregnancy:

1- Smoking: heavy smoking is associated with IUGR even IUFD

* Should be encourage to stop smoking

2- Rest: 8 hr at night & 1 – 2 hr after noon.

3- Sexual intercourse: Not in 1st trimester ⟶ to ↓ incidence of abortion. Not in late month to avoid premature rupture of membrane

4- Clothes: comfortable avoid high heel shoes.

II- Diet system: 1-Total caloric intake /d = 2400kc.

2-↑ Protein intake especially of animal origin.

3-Carbohydrate is the same amount as in between pregnancy.

4- Supplementation.

Iron: iron supplementation especially after 12 w of pregnancy.

Folic acid: especially in first trimester.

Calcium: sometimes especially late in pregnancy.

5- ↑ Fluid intake in diet to avoid constipation.

III-Instruct patient with minor disturbances with pregnancy: e.g. ptyalism, heart burn, ↑ vaginal discharge, leg cramps.

IV-Instruct patient with major disturbances (warning signs) of pregnancy: e.g any vaginal bleeding, gush of water (PROM),abdominal pain, continous headache, generalized oedema, blurred vision, persist vomiting, fever with rigors, dysuria

V- Preparation for lactation.

(B) Detection & correction of any abnormalities:

This will be conducted by visits.

** Frequency of visits:*

- -Low risk pregnancy: every month for 7 months then / 2 weeks for 2 months then/ week from 36 – 40 w.
- High risk pregnancy: every 2 weeks up to 28 w then / week from 28-36 w, then hospitalization from 36 w till delivery.

**** 1st visit (Booking visit):***

- Complete history taking: including personal, present, past, family, menstrual, obstetric & social history.
- Complete clinical exam: general , abdominal, vaginal exam.
- Investigation: A) Routine investigation:

  1- Urine analysis: for proteins, glycosuria,microscopic exam. for pus.

  2- Blood exam. For: Hb% ,MCV of RBC,ABO &Rh.

  3-Blood glucose level .

  B) Other tests: 1-Hepatitis β-surface A.g.

  2- Cervical & urinary culture.

  3- Toxoplasmosis antibodies.

  4- Rubella antibodies.

  C) Use of image technique: U/S for pregnant uterus.

**** Subsequent ANC-visits:***

* At all visits, the following items should be checked:

weight gain, A.B.P, fundal level, auscultation of F.H.R.

N.B: Nutrition (diet) with normal pregnancy:

1- So with pregnancy = 2500 Kcal. /day.

2- Major food staff CHO = 300 gm /d.

Protein = 90 – 100 gm/d.

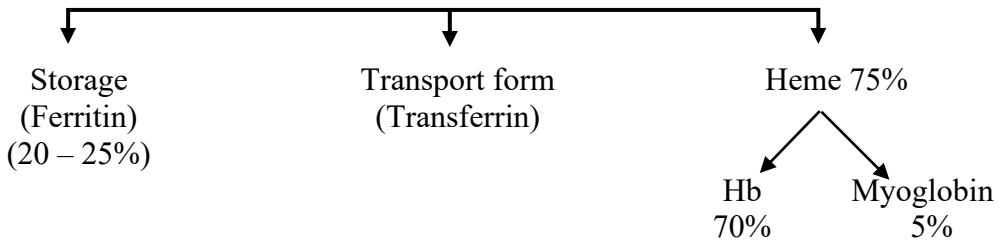
Lipids = 90 – 100 gm /d.

3- Minerals:

a) Iron :

- body iron : the total iron in male body = 4 gm.

In adult young ♀ (2 – 2.5 gm).



- Iron requirement with pregnancy:

Fetus + placenta → 300 mg.

Expansion of blood volume → 500 mg.

Daily loss 1 mg/d. x gestational period = 250 mg.

So the total iron need all over pregnancy = 1 gm.

1000 mg/ 250 day of pregnancy = 4 mg/d.

As iron absorption is about 10 – 15% of total elementary iron intake so every ♀ pregnancy need 30 – 40 elementary iron /d.

Take care that:

a) Iron requirement is usually in second half of pregnancy As in 1st Tr. iron level nearly the same as non pregnancy 60 – 120 ug% .

b) Iron can cause nausea and vomiting.

c) Iron requirement return to non – pregnancy need during lactation.

b) Ca^{++} :

- Body Ca^{++}: The total ♀ body Ca = 1200 gm, most of them in skeleton.

- Ca^{++} requirement with pregnancy: total need about 30 gm. mainly to fetus especially late in pregnancy

The daily in take of 1.2 gm Ca/d is enough which may be replaced by milk and milk products.

c) Phosphorus: - Need as Ca^{++}, 1.2 gm /d.

d) Zinc: Deficiency may lead to

- Poor appetite
- Impaired healing.
- PROM
- IUGR.

The total daily intake with pregnancy = 15 mg /d which present in balanced diet.

e) Iodine: ↓ may lead to goiter & critinism.

f) Fluoride : ↓ may lead to dental caries.

g)Selenium: Which is essential compound of enzyme glutathione peroxidase which catalyse hydrogen peroxide → H_2O and avoid appear of free radicals which can cause multiple-opathies e.g. cardiomyopathies.

h) Chromium: Important for action of insulin facilitate attachment of insulin to peripheral receptors.

Vitamins	Deficiency	Recommendation
- A	Dry skin, ↓ epith.	6000 IU/d.
- D	Osteomalacia (Mother) & Rickets (baby).	600 IU/d.
- C	PPH, scurvey.	60 mg/d.
- B. complex	Vomiting, beri beri.	30 mg/d.
-Folic acid	Anemia, NTD, abortion, PTL, IUGR, IUFD, other cong. anomalies	1 mg/d.

N.B. : How to estimate fetal weight (clinically):

Johnson's formula.

Weight (in gm) = (fundal height (cm) – n) x 155

N = 12 If vertex above ischial spine

Or = 11 If vertex below ischial spine .

} For cephalic presentation